NPTEL ONLINE CERTIFICATION COURSES




Course Name: Software Project Management

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Department : Computer Sc. & Engg. NIT Rourkela

TOPIC: Contract management




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## CONCEPTS COVERED

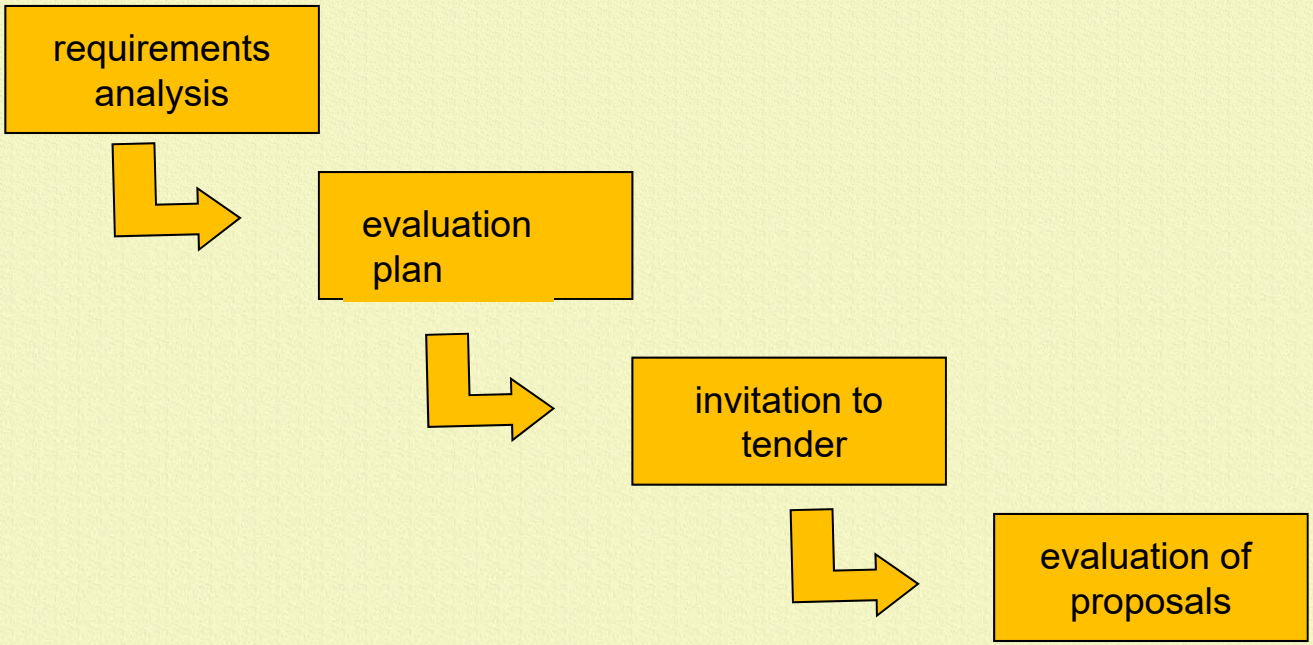
- Stages in contract placement
- Contract checklist
- Contract management




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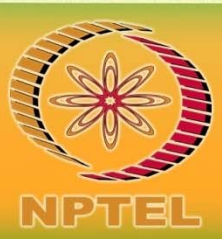
# Stages in contract placement



## Requirements document: sections

- Introduction
- Description of existing system and current environment
- Future strategy or plans
- System requirements -
  - mandatory/desirable features
- Deadlines
- Additional information required from bidders

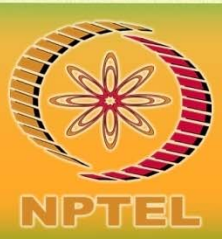


# Requirements

- These will include
  - functions in software, with necessary inputs and outputs
  - standards to be adhered to
  - other applications with which software is to be compatible
  - quality requirements e.g. response times

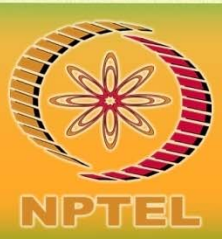


# Evaluation plan

- How are proposals to be evaluated?
- Methods could include:
  - reading proposals
  - interviews
  - demonstrations
  - site visits
  - practical tests

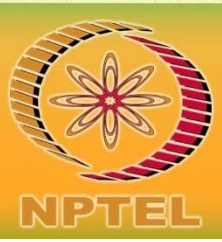


# Evaluation plan -contd.

- Need to assess value for money (VFM) for each desirable feature
- VFM approach is an improvement on previous emphasis on accepting lowest bid
- Example:
  - feeder file saves data input
  - 4 hours work a month saved at £20 an hour
  - system to be used for 4 years
  - if cost of feature £1000, would it be worth it?

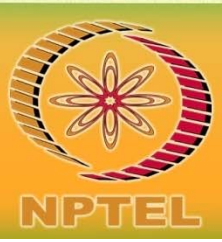


# Invitation to tender (ITT)

- Note that bidder is making an *offer* in response to ITT
- *acceptance* of offer creates a *contract*
- Customer may need further information
- Cons: Different technical solutions to the same problem, so difficult to evaluate

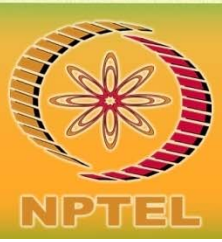


# Memoranda of agreement (MoA)

- Customer asks for technical proposals
- Technical proposals are examined and discussed
- Agreed technical solution is MoA
- Tenders are then requested from suppliers based on MoA
- Tenders are judged on price
- Fee could be paid for technical proposals by customer

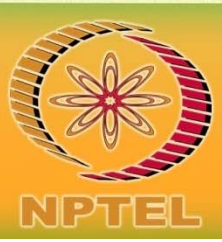


## Evaluation of proposals

- It is needed to produce an evaluation plan describing how each proposal will be checked against the selection criteria.
- This reduces risks of requirements being missed and ensures that all proposals are treated consistently.
- It would be unfair to favour a proposal because of the presence of a feature not requested in the original requirement.

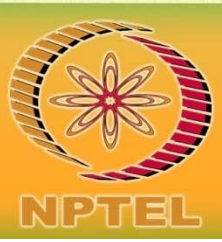


# Evaluation of proposals cont...

- An application could be bespoke, off-the-shelf, or customized.
- In the case of off-the-shelf packages, the software itself could be evaluated and it might be possible to combine some of the evaluation with acceptance testing.
- With bespoke development, it would be a proposal that is evaluated, while COTS may involve elements of both. Thus different approaches would be needed.



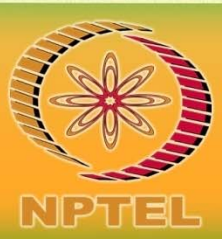



# Evaluation of proposals cont...

The process of evaluation may include:

- Scrutiny of the proposal documents
- Interviewing suppliers' representatives
- Demonstrations
- Site visits
- Practical tests

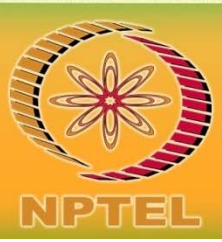


# Evaluation of proposals cont...

- The proposal documents provided by the suppliers can be scrutinized to see if they contain features satisfying all the original requirements.
- Clarification might be sought over certain points. It is important to get a written, agreed, record of these clarifications.
- The customer might take initiative by taking minutes of meetings and then writing afterwards to the suppliers to get them to confirm their accuracy.
- A supplier could, in the final contract document, attempt to exclude any commitment to any representations made in pre-contract negotiations – the terms of the contract need to be scrutinized for this. Where there is an existing product, there could be a demonstration.

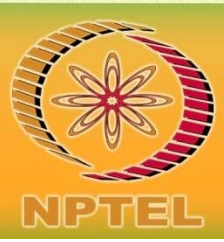


## Evaluation of proposals cont...

- A frequent problem is that while an existing application works well on one platform with a certain level of transactions, it does not work satisfactorily with the customer's ICT configuration or level of throughput.
- Demonstrations might not reveal this problem. Visits to operational sites already using the system, could be more informative.
- In the last resort a special volume test could be conducted.

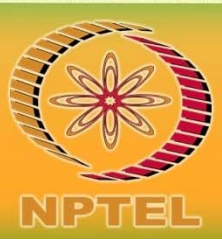


# Evaluation of proposals cont...

- Finally, a decision is made to award the contract to a supplier.
- The reason for following a structured and objective approach to evaluation is to demonstrate that the decision has been made impartially.
- A project manager cannot be expected to be a legal expert – needs advice.
- BUT must ensure that the contract reflects true requirements and expectations of supplier and client.



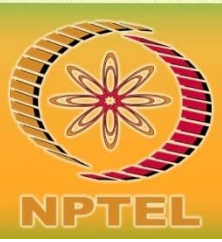



# Contract checklist

- Definitions – what words mean precisely e.g. ‘supplier’, ‘user’, ‘application’

- Form of agreement - For example, is this a contract for a sale or a lease, or a license to use a software application? Can the license be transferred?



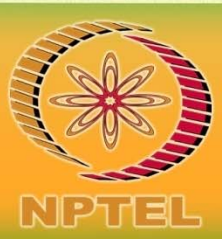



# Contract checklist

Goods and services to be supplied –

- *Equipment and software to be supplied:* this should include an actual list of the individual pieces of equipment to be delivered, complete with the specific model numbers.
- *Services to be provided:* This would cover such things as
  - Training
  - Documentation
  - Installation
  - Conversion of existing files
  - Maintenance agreements
  - Transitional insurance arrangements

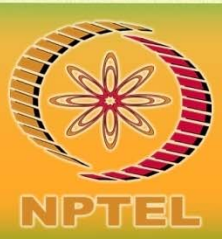


# Contract checklist - continued

- Ownership of software
  - Can client sell software to others?
  - Can supplier sell software to others? Could specify that customer has ‘exclusive use’
  - Does supplier retain the copyright?
  - Where supplier retains source code, may be a problem if supplier goes out of business; to circumvent a copy of code could be deposited with an **escrow** service



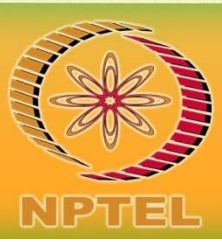
# Contract checklist - continued

- Environment – for example, where equipment is to be installed, who is responsible for various aspects of site preparation e.g. electricity supply, back up facilitation etc.?
- Customer commitments – for example providing access, supplying information

# Contract checklist - continued

- Acceptance procedures – Good practice is to accept a delivered system only after user acceptance tests.
- Part of the contract would specify such details as the time that the customer will have to conduct the tests, deliverables upon which the acceptance tests depend and the procedure for signing off the testing as completed.

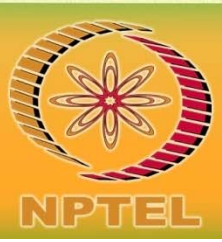


# Contract checklist - continued

- Standards to be met – This covers the standards with which the goods and services should comply.
- For example, a customer could require the supplier to conform to the ISO 12207 standard relating to the software life cycle and its documentation (or, more likely, a customized sub-set of the standard).

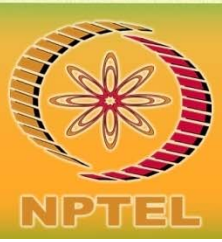


## Contract checklist

- Project and quality management – the arrangements for the management of the project must be agreed. These include the frequency and nature of progress meetings and the progress information to be supplied to the customer.
- Timetable of activities – This provides a schedule of when the key parts of the project should be completed



# Contract checklist

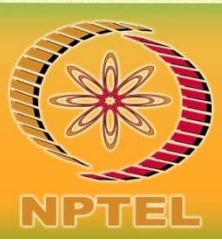
- Price and payment method– obviously the price is very important. what also needs to be arranged is when the payments are to be made.
- payments may be tied to completion of specific tasks.

# Contract management

Some terms of contract will relate to management of contract, for example,

- Progress reporting
- Decision points – could be linked to release of payments to the contractor
- Variations to the contract, i.e. how are changes to requirements dealt with?
- Acceptance criteria

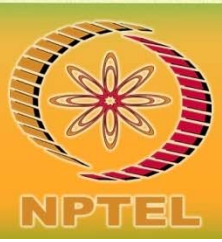


## How would you evaluate the following?

- usability of an existing package
- usability of an application yet to be built
- maintenance costs of hardware
- time taken to respond to requests for software support
- training

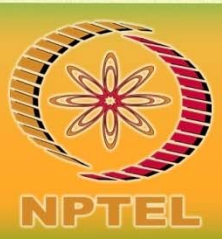


# Contract management cont …

- Contracts should include agreement about how customer/supplier relationship is to be managed e.g.
  - *decision points* - could be linked to payment
  - *quality reviews*
  - *changes to requirements*

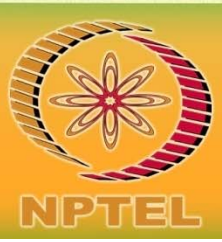


# Summary

- Explained the different stages in contract placement
- Discussed the typical terms of a contract
- Explained some terms of contract which relate to management of contract

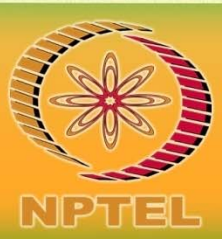


# References :

1. B. Hughes, M. Cotterell, R. Mall, *Software Project Management*, Sixth Edition, McGraw Hill Education (India) Pvt. Ltd., 2018.




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